Module - I : Reactions Kinetics

Rate of Reaction:

The small change in concentration per unit time is called rate of reaction.

$$\underset{\text{(Reactant)}}{A} \longrightarrow \text{Product}$$

Initially	a	0
t-time	$(a-x)$	x

$$\text{Rate} = \frac{dx}{dt}$$

Law of Mass Action:

This states that, "The rate of reaction is directly proportional to reactant concentration."

$$\text{i.e. Rate} \propto [\text{Reactant}]$$

$$\frac{dx}{dt} \propto [A]$$

$$\frac{dx}{dt} = K[A]$$

where, 'K' is rate constant.

Examples;

(i) $H_2 + I_2 \longrightarrow 2HI$

$$Rate = K[H_2][I_2]$$

(ii) $2SO_2 + O_2 \longrightarrow 2SO_3$

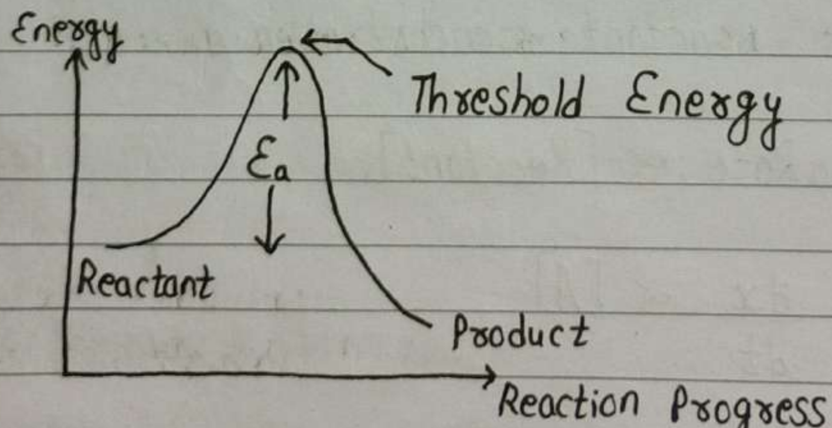
$$Rate = K[SO_2]^2[O_2]$$

Factors Affecting Rate of Reaction

① Effect of concentration: Greater the concentration of reactant, greater will be the inter-molecular collisons and hence greater will be the rate of reaction.

② Effect of catalyst: Catalyst increase the rate of reaction by lowering the activation energy (E_a).

$$E_a = E_{threshold} - E_{reactant}$$



③ Effect of Temperature: With an increase of 10°C, the rate of reaction approximately doubles.

$$\phi = \frac{K_{T+10}}{K_T}$$

where, ϕ is temperature coefficient and is defined as the ratio of the rate constants of a reaction measured at two temperatures that differ by 10°C.

'T' is the temperature.

Molecularity

Molecularity is defined as the number of reactant species colliding in an elementary reaction.

It cannot be zero, fractional and does not exceed 4.

Example;

$$H_2 + I_2 \longrightarrow 2HI \quad [\text{Molecularity} = 2]$$

Order of Reaction

Order of reaction is the sum of the exponents of the concentration terms in a reaction's rate law. It is determined experimentally. It can be 0, 1, 2, 3.

$$aA + bB \longrightarrow \text{Products}$$

$$\text{Rate} = K[A]^a[B]^b$$

$$\text{Order} = a + b.$$

For example;

(i) $2SO_2 + O_2 \longrightarrow 2SO_3$

$$Rate = K[SO_2]^2[O_2]^1$$

Order = 2+1 = 3 (3^{rd} order reaction)

(ii) $CH_3COOC_2H_5 + H_2O \longrightarrow CH_3COOH + C_2H_5OH$

$$Rate = K[CH_3COOC_2H_5]^1[H_2O]$$

Order = 1 (Pseudo 1^{st} order reaction)

Note: $[H_2O]$ is constant.

Note: Pseudo first order reaction is a reaction that is actually of higher order but appears to be first order because one reactant is present in large excess (conc. stays constant).

There are following types of order of reaction:

1. Zero Order Reaction
2. First Order Reaction
3. Second Order Reaction
4. Third Order Reaction

Another Definition: The sum of the power of reactant moles is called order of reaction.

Order of Reaction	Molecularity
(i) It is sum of the exponents to which the conc. terms in rate law expression are raised.	It is the number of molecules of the reactants taking part in the elementary reaction.
(ii) It is experimentally determined and indicated the dependence of the reaction rate on the conc. of particular reactants and hence it depends one experimental conditions.	It is theoretical property and indicates the no. of molecules of reactant in each step of the reaction and hence it does not depends on experimental conditions.
(iii) It may have values which are integer, fractional or zero.	It is always an integer.
(iv) It is the property of elementary and complex reactions.	It is the property of elementary reactions only.
(v) Rate law expression describes the order of the reaction.	Rate law does not describe molecularity.

For example;

(i) $PCl_5 \longrightarrow PCl_3 + Cl_2$

Rate $= K[PCl_5]^1$

Molecularity $= 1$

Order of reaction $= 1^{st}$ order

(ii) $2NO + O_2 \longrightarrow 2NO_2$

Rate $= K[NO]^2[O_2]^1$

$M = 3$

Order of reaction $= 3^{rd}$ order

Exceptional Example;

(i) $H_2 + Cl_2 \xrightarrow{h\nu} 2HCl$

Rate $= K[H_2]^0[Cl_2]^0$

$M = 2$

Order of reaction = zero order

(ii) ~~CH3~~ $C_{12}H_{22}O_{11} + H_2O \xrightarrow{H^+} C_6H_{12}O_6 + C_6H_{12}O_6$

Rate $= K[C_{12}H_{22}O_{11}]^1[H_2O]$

$M = 2$

Order of reaction = Pseudo-first order.

(iii) $2NH_3 \xrightarrow[\Delta]{Pt/Au} N_2 + 3H_2$

$Rate = k[NH_3]^0$

$M = 2$

Order of reaction = zero order.

Unit of Rate Constant

$$K = \text{Mole lit}^{-1}\ \text{sec}^{-1} \times \frac{1}{(\text{Mole lit}^{-1})^n}$$

For $n = 1$,

$$K = \cancel{\text{Mole lit}^{-1}}\ \text{sec}^{-1} \times \frac{1}{\cancel{(\text{Mole lit}^{-1})^1}}$$

$$\boxed{k = \text{sec}^{-1}}$$

For $n = 0$,

$$k = \text{Mole lit}^{-1}\ \text{sec}^{-1}$$

For $n = 2$,

$$k = \text{Mole}^{-1}\ \text{lit}\ \text{sec}^{-1}.$$

For $n = 3$,

$$k = \text{Mole}^{-2}\ \text{lit}^2\ \text{sec}^{-1}$$

Zero Order Reaction: A chemical reaction where the rate of reaction is independent of the concentration of the reactants is known as zero-order reaction.

$$A \longrightarrow \text{Product}$$

Initially	a	0
t-time	$(a-x)$	x

According to law of mass action,

$$\text{Rate} \propto [A]^0 \quad [\text{For zero-order reaction}]$$

or, $\dfrac{dx}{dt} = K(a-x)^0$

or, $\dfrac{dx}{dt} = K$

or, $dx = K\,dt$

Integrating both sides,

$$\int dx = K \int dt$$

or, $x = Kt + I$, where 'I' is Integration constant.

At $t=0$, $x=0$;

$$0 = K \times 0 + I$$

$$\Rightarrow I = 0$$

So, the final eq^n. is,

$$x = Kt$$

$$\Rightarrow \boxed{K = \frac{x}{t}}$$

This is integrated rate law for zero-order reaction.

The unit of K for zero-order reaction is Mole $lit^{-1}s^{-1}$.

Half-life for zero-order reaction: Half-life means the time it takes for the concentration of the reactant to reduce to half of its original value.

At $t = T_{1/2}$ & $x = a/2$,

$$K = \frac{a}{2T_{1/2}}$$

i.e. $\boxed{T_{1/2} = \frac{a}{2K}}$

For example,

(i) $H_2 + Cl_2 \xrightarrow{h\nu} 2HCl$

Rate $= K[H_2]^0[Cl_2]^0$

(ii) $2NH_3 \xrightarrow[\Delta]{Au/Pt} N_2 + 3H_2$

Rate $= K[NH_3]^0$.

First Order Reaction:

$$A \longrightarrow \text{Product}$$

$t=0$	a	0
t time	$(a-x)$	x

From Law of Mass Action for $n=1$,

$$\frac{dx}{dt} \propto [A]^1$$

or, $$\frac{dx}{dt} = K(a-x)$$

or, $$\frac{dx}{(a-x)} = K\,dt$$

On Integration,

$$\int \frac{dx}{(a-x)} = K \int dt$$

or, $$-\log_e(a-x) = Kt + I \quad —(1)$$

At, $t=0$ & $x=0$,

$$-\log_e(a-0) = K \times 0 + I$$

or, $$I = -\log_e a$$

Eq^n. (1) becomes,

$$-\log_e(a-x) = Kt - \log_e a$$

or, $kt = \log_e a - \log_e (a-x)$

or, $kt = \log_e a \dfrac{a}{(a-x)}$

or, $kt = 2.303 \log_{10} \dfrac{a}{(a-x)}$

$$\boxed{k = \frac{2.303}{t} \log_{10} \frac{a}{(a-x)}}$$

This is integrated rate law for first order reaction.

The unit of k is sec^{-1}.

Half-life

$t = T_{1/2}$ & $x = a/2$

$t = \dfrac{2.303}{k} \log_{10} \dfrac{a}{(a-x)}$

or, $T_{1/2} = \dfrac{2.303}{k} \log_{10} \dfrac{a}{(a-a/2)}$

or, $T_{1/2} = \dfrac{2.303}{k} \log_{10} \dfrac{2\cancel{a}}{\cancel{a}}$

or, $T_{1/2} = \dfrac{2.303}{k} \log_{10} 2$

or, $T_{1/2} = \frac{2.303}{K} \times 0.3010$

$\therefore \boxed{T_{1/2} = \frac{0.693}{K}}$

For example;

(i) $H_2O_2 \longrightarrow H_2O + 1/2\ O_2$

Rate $= K[H_2O_2]^1$

(ii) $PCl_5 \longrightarrow PCl_3 + Cl_2$

Rate $= K[PCl_5]^1$

(iii) $N_2O_5 \longrightarrow 2NO_2 + 1/2\ O_2$

Rate $= K[N_2O_5]^1$

Question - First Type ~~(OR section)~~

Q) For a first order reaction, calculate the ratio between the time taken to complete three-fourth of the reaction and the time taken to complete half of the reaction?. Also 75 % of reaction is twice of 50 %?

Soln:

~~Given,~~ $\frac{t_{3/4}}{t_{1/2}} = ?$

& $t_{75} = 2 \times t_{50}$

For 75%,

$a = 100$ & $x = 75$

Then, $K = \frac{2.303}{t} \log_{10} \frac{a}{(a-x)}$

or, $t_{75} = \frac{2.303}{K} \log_{10} \frac{100}{100-75}$

or, $t_{75} = \frac{2.303}{K} \log_{10} 2^2$

or, $t_{75} = \frac{2.303}{K} \times 2 \times 0.3010$ —— (1)

For 50%,

$a = 100$ & $x = 50$

~~t_{50}~~ $t_{50} = \frac{2.303}{K} \log_{10} \frac{\cancel{100}\,2}{\cancel{50}}$

or, $t_{50} = \frac{2.303 \times 0.3010}{K}$ — (2)

$\frac{t_{75}}{t_{50}} = \frac{2.303 \times 2 \times 0.3010 / K}{2.303 \times 0.3010 / K}$

$\therefore \frac{t_{75}}{t_{50}} = 2$

i.e. $t_{75} = 2 \times t_{50}$.

Question - Second Type

Q) A first-order reaction, which is completed 90% in 10 hrs. When will 99.9% of the reaction is complete?

Soln:

90%

$a = 100$ & $x = 90$ $t = 10$ hrs

$$K = \frac{2.303}{t} \log_{10} \frac{a}{(a-x)}$$

$$= \frac{2.303}{10} \log_{10} \frac{\cancel{100}\,10}{\cancel{10}}$$

$$= \frac{2.303}{10} \text{ sec}^{-1}.$$

99.9%

$a = 100$, $x = 99.9$ & $t = ?$

$$t = \frac{2.303}{k} \log_{10} \frac{a}{(a-x)}$$

$$= \frac{2.303}{2.303/10} \log_{10} \frac{100}{0.1}$$

$$= 10 \times \log_{10}(1000)$$

$$= 30$$

$\therefore$ 30 hrs time is taken to complete the 99.9% of reaction.

Q) A first-order reaction is 20 % complete in 10 minutes. Calculate the time taken for the reaction go to 80 % completion?

Soln:

<u>20 %</u>

$a = 100\%$, $x = 20\%$ & $t = 10$ min

$$K = \frac{2.303}{t} \log_{10}\left(\frac{a}{a-x}\right)$$

$$= \frac{2.303}{10} \log_{10}\left(\frac{100}{80}\right)$$

$$= \frac{2.303}{10} \log_{10}\left(\frac{5}{4}\right)$$

$$\therefore K = 0.022318 \ sec^{-1}$$

<u>80 %</u>

$a = 100\%$, $x = 80\%$ & $t = ?$

$$K = \frac{2.303}{t} \log_{10}\left(\frac{a}{a-x}\right)$$

$$\therefore t = \frac{2.303}{0.022318} \log_{10}\left(\frac{\overset{5}{\cancel{100}}}{\cancel{20}}\right)$$

$$= 72.12 \text{ minutes.}$$

Imp.

Second Order Reaction:

(i) When both reactants are same;

$$A + A \longrightarrow \text{Product}$$

$$\text{Rate} = K[A]^2$$

(ii) When both reactants are different;

$$A + B \longrightarrow \text{Product}$$

$$\text{Rate} = K[A]^1[B]^1$$

(i) When both reactants are same

	$2A$	$\longrightarrow$ Product
Initially	a	0
t-time	$(a-x)$	x

Law of Mass Action,

$$\frac{dx}{dt} \propto [A]^2$$

or, $$\frac{dx}{dt} = K(a-x)^2$$

or, $$\frac{dx}{(a-x)^2} = K\,dt$$

On Integration,

$$\int \frac{dx}{(a-x)^2} = K \int dt$$

or, $\frac{1}{(a-x)} = kt + I$ — (1)

At $t=0$, $x=0$,

$$\frac{1}{(a-0)} = k\times 0 + I$$

$$I = \frac{1}{a}$$

Eqn. (1) becomes,

$$\frac{1}{(a-x)} = kt + \frac{1}{a}$$

$$kt = \frac{1}{(a-x)} - \frac{1}{a}$$

$$kt = \frac{x}{a(a-x)}$$

$$\therefore \boxed{k = \frac{1}{t}\,\frac{x}{a(a-x)}}$$

This is integrated rate law for second order reaction when both conc. are same.

Imp

(ii) When both reactants are different

$$A + B \longrightarrow \text{Product}$$

Initially,	'a'	'b'	0
t-time,	$(a-x)$	$(b-x)$	x

Law of mass-action,

$$\frac{dx}{dt} = K[A]^1[B]^1$$

or, $$\frac{dx}{dt} = K(a-x)(b-x)$$

or, $$\frac{dx}{(a-x)(b-x)} = K\,dt$$

Spliting into partial fraction,

$$\frac{1}{(a-b)}\left[\frac{dx}{(b-x)} - \frac{dx}{(a-x)}\right] = Kdt$$

On Integration,

$$\frac{1}{(a-b)}\left[-\log_e(b-x) + \log_e(a-x)\right] = Kt + I \quad \text{—— (1)}$$

At, $t=0$ & $x=0$,

$$\frac{1}{(a-b)}\left[\log_e(a-0) - \log_e(b-0)\right] = K \times 0 + I$$

i.e. $$I = \frac{1}{(a-b)}\log_e\left(\frac{a}{b}\right)$$

Eq^n. (1) becomes,

$$\frac{1}{(a-b)} \log_e\left(\frac{a-x}{b-x}\right) = Kt + \frac{1}{(a-b)} \log_e\left(\frac{a}{b}\right)$$

$$\therefore \boxed{K = \frac{1}{t(a-b)} \log_e \frac{b(a-x)}{a(b-x)}}$$

Half-life, $T_{1/2} \propto \frac{1}{a}$

Pseudo-Order Reaction:

In this reaction two are reactants are involved in which one reactant in excess conc., this type of reaction is known as ~~ps~~ pseudo order reaction.

For example,

(i) $CH_3COOC_2H_5 + H_2O \xrightarrow{H^+} CH_3COOH + C_2H_5OH$
Ethyl Acetate — Acetic Acid — Ethanol

$$Rate = K[CH_3COOC_2H_5]^1 \underset{Excess}{[H_2O]}$$

(ii) Pseudo order reaction,

$C_{12}H_{22}O_{11} + H_2O \xrightarrow{H^+} C_6H_{12}O_6 + C_6H_{12}O_6$
Sucrose — Glucose — Fructose

$$Rate = K[C_{12}H_{22}O_{11}]^1 [H_2O] \text{ (Excess)}$$

Arrhenious Equation To Calculating Activation Energy of the Reaction

$$k = A \cdot e^{-E_a/RT}$$

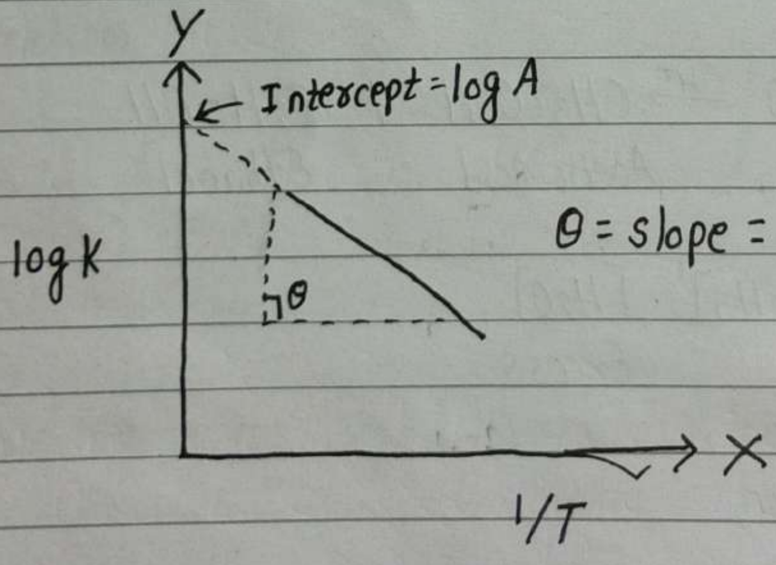
- k → Rate constant
- A → Arrhenious constant
- E_a → Activation Energy
- R → Gas Constant
- T → Absolute Temperature (K)

$$\boxed{R = 8.314 \ JK^{-1} mol^{-1}}$$

Graphical Representation

$$\log k = \log A - \frac{E_a}{2.303RT}$$

$$\text{or, } \log k = \frac{-E_a}{2.303R}\frac{1}{T} + \log A \qquad [y = mx + c]$$



$$\theta = \text{slope} = \frac{-E_a}{2.303R}$$

Fig: log k Vs 1/T

Calculation of Activation Energy (E_a)

$$K = A . e^{-E_a/RT}$$

$$\log K = \frac{-E_a}{2.303R} \frac{1}{T} + \log A$$

For first-order reaction, rate constant (K_1)

$$\log K_1 = \frac{-E_a}{2.303R} \frac{1}{T_1} \quad \text{— (1)}$$

For second-order reaction, rate constant (K_2)

$$\log K_2 = \frac{-E_a}{2.303R} \frac{1}{T_2} \quad \text{— (2)}$$

(2) - (1),

$$\log K_2 - \log K_1 = \frac{-E_a}{2.303R} \left[\frac{1}{T_2} - \frac{1}{T_1}\right]$$

$$\log \frac{K_2}{K_1} = \frac{E_a}{2.303R} \left[\frac{1}{T_1} - \frac{1}{T_2}\right]$$

$$\boxed{\log \frac{K_2}{K_1} = \frac{E_a}{2.303R} \left[\frac{T_2 - T_1}{T_1 T_2}\right]}$$

Q) The rate of reaction triples when the temperature changes from 20°C to 50°C. Calculate activation energy.

Soln:

Given, $\frac{k_2}{k_1} = 3$

$R = 8.314 \text{ J K}^{-1} \text{ mol}^{-1}$

$T_1 = 20°C = 293 \text{ K}$

$T_2 = 50°C = 323 \text{ K}$

We know,

$$\log \frac{k_2}{k_1} = \frac{E_a}{2.303 R} \left[\frac{T_2 - T_1}{T_2 T_1} \right]$$

$$\text{or, } \log 3 = \frac{E_a}{2.303 \times 8.314} \left[\frac{323 - 293}{323 \times 293} \right]$$

$$\text{or, } \frac{E_a}{19.147} \times \frac{30}{94639} = 0.477$$

$\therefore E_a = 28811.64 \text{ J/mol}$

i.e. $E_a = 28.8 \text{ KJ/mol}$ ##

Q) The rate of reaction doubles when the temperature changes from 20°C to 50°C. Calculate the energy of activation of the reaction. [$R = 8.314 \text{ J K}^{-1} \text{mol}^{-1}$]

Soln:

$k_2 = 2k_1$

$R = 8.314 \text{ J K}^{-1} \text{mol}^{-1}$

$T_1 = 20°C = 293\ K$
$T_2 = 50°C = 323\ K$
$E_a = ?$

We know,

$$\log \frac{K_2}{K_1} = \frac{E_a}{2.303\ R}\left[\frac{T_2 - T_1}{T_1 T_2}\right]$$

or, $$\log 2 = \frac{E_a}{2.303 \times 8.314}\left[\frac{323 - 293}{293 \times 323}\right]$$

or, $$0.3010 = \frac{E_a}{19.14} \times \frac{30}{94639}$$

$\therefore E_a = 18,174.28\ J/mol$
i.e. $E_a = 18.17\ kJ\ mol^{-1}$.

Q) The rate of a reaction quadruples when the temperature changes from 293 to 313 K. Calculate the energy of activation of the reaction.

Sol^n:

$K_2 = 4K_1$
$R = 8.314\ J K^{-1} mol^{-1}$
$T_1 = 293\ K$
$T_2 = 313\ K$
$E_a = ?$

$$\log \frac{K_2}{K_1} = \frac{E_a}{2.303\ R}\left[\frac{T_2 - T_1}{T_1 T_2}\right]$$

or, $\log 4 = \frac{E_a}{2.303 \times 8.314}\left[\frac{313-293}{313 \times 293}\right]$

$\therefore E_a = 52,859.82 \ J\ mol^{-1}$
i.e. $E_a = 52.85 \ kJ\ mol^{-1}$.

Theory of Reaction Rates

1. Collision Theory:
A chemical reaction occurs only as a result of collisions between the reacting species, during such type of collisions, atoms get rearranged by breaking ~~of~~ old bonds and forming new bonds.

For example;

$$A_2(g) + B_2(g) \longrightarrow 2AB(g)$$

Formation of AB is only possible if molecules A_2 and B_2 collide.

Collision Frequency (Z): The total number of collisions that occur among the reacting species per second per unit volume is called collision frequency (Z).

Effective Collision: The collisons that bring about the formation of product are called effective collisions.

For a reaction to occur, their are two important barriers:

(i) <u>Energy Barrier</u>: For collisions, a certain minimum amount of energy is required. The minimum energy which colliding particles must possess in order to bring about product formation is called Threshold Energy.

(ii) <u>Orientation Barrier</u>: All molecules having higher energy than threshold energy may not always lead to reaction. Proper orientation during collision is also required.

For a chemical reaction:

(i) Reactant must collide with each other.
(ii) They must have proper orientation.
(iii) Energy must be equal to or greater than threshold energy.

<u>Reaction Rate Expression</u>:

The specific reaction rate 'K' is proportional to the number of effective collisions per second.

$$\text{Rate of reaction} = f \times Z$$

where, f = fraction of collisions that are effective.
Z = collision frequency.

So, $K = Z \times f$

And,

$$k = Z \times e^{-E_a/RT}$$

This eq^n. is smilar to the Arrhenius equation:

$$k = A e^{-E_a/RT}$$

where, $A = Z$.

2. Transition State Theory (Activated Complex Theory):
Collisions of molecules having energy less than threshold energy do not form products. It means that between reactants and products, there exists an energy barrier that must be crossed.

The extra energy that reactant molecules require to attain threshold energy in order to yield products is called Activation Energy (E_a).

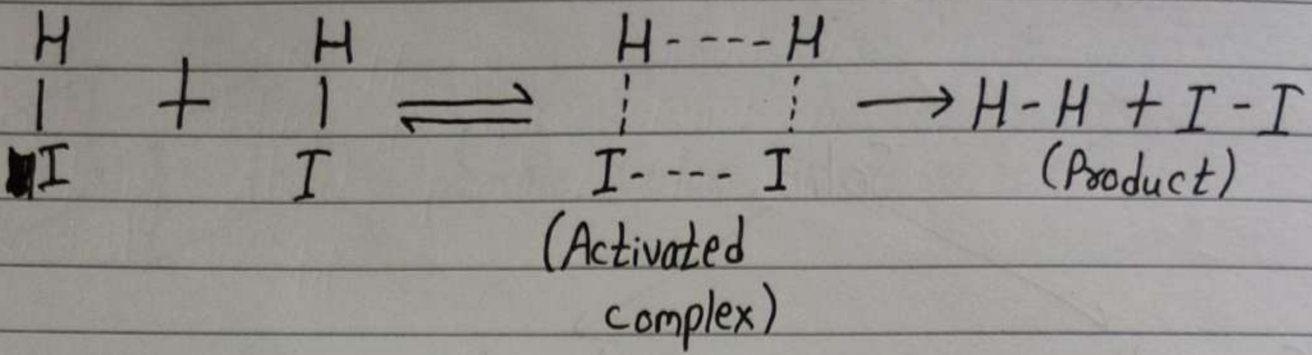
Activation Energy = Threshold Energy − Energy possessed by reactant

Reactant → Activated Complex ⟶ Products

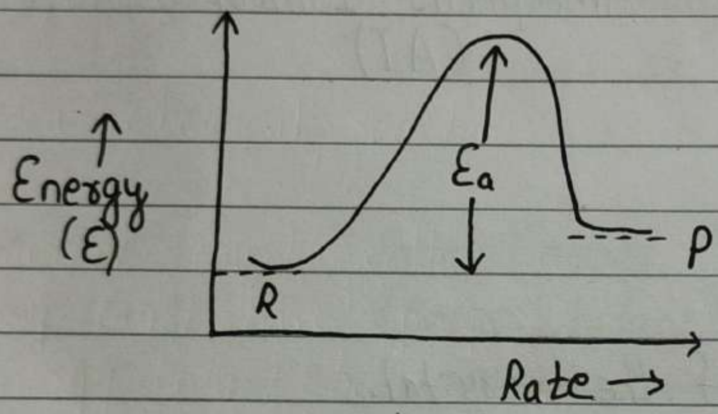
$$A + B = [X]^* \longrightarrow \text{Product}$$

For example;

$$H_2 + I_2 \longrightarrow 2HI$$



If $\mathcal{E}_a$ is high, reaction is slow.
If $\mathcal{E}_a$ is low, reaction is fast.



Rate Equation:

$$k = A e^{-\mathcal{E}_a/RT}$$

where, A is the frequency factor, $\mathcal{E}_a$ is activation energy, R is gas constant & T is temperature.